Raynaud's theorem

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The material follows the lecture notes in [Sno13].

The Raynaud's theorem is to say in mixed characteristic and low ramification field, a group scheme is determined by its generic fiber.

1 Setting up

Let $K/\mathbb{Q}_p$ be a finite extension, $G_0 := \operatorname{Spec} A_0$ be a commutative finite group scheme over K, let e be ramification index of K, p be a prime, and $R := O_K$ is rings of integer of K. Let A_0 be a finite K-Hopf-algebra.

Definition 1.1. A *prolongation* of group scheme G_0 is a finite flat group scheme $G := \operatorname{Spec} A$ over R such that $G_K := G \times_{\operatorname{Spec} R} \operatorname{Spec} K \cong G_0$.

Precisely, the prolongation of G_0 corresponding to the sub R-algebra $A \subset A_0$ that span the A_0 over K and also satisfies the Hopf-algebra axioms:

- comultiplication: $m^* : A \longrightarrow A \otimes_R A$,
- antipode : $i^* : A \xrightarrow{\sim} A$,
- counit: $e^* : A \to R$.

We can order the prolongations using inclusion of rings.
The Cartier dual of G $G^\vee := \underline{\operatorname{Hom}(G, \mathbb{G}_m)}$ has algebra $\operatorname{Hom}_R(A, R) = B$.
We say G has UP means G has unique prolongation.

Theorem 1.2 (Raynaud)**.** *Every finite flat group scheme over K has UP if $e < p - 1$.*

Remark 1.3. The hypothesis $e < p - 1$, is very important. If K has p-th roots of unity, then $(\mu_p)_R$ and constant group scheme $\underline{\mathbb{Z}/p\mathbb{Z}}_R$ both are prolongations of $\underline{\mathbb{Z}/p\mathbb{Z}}_K$, but they are not isomorphic, since on special fiber, one is etale, the other is connected.

The idea of the proof separates into the following 3 steps:

- If G has UP, then its subgroups and also the quotient of G have UP $\implies$ simple group G has UP.
- Classify the simple group G
- Verify the condition $e < p - 1$ which indeed implies the simple G has UP.

2 Analysis of prolongation

Theorem 2.1. *The prolongation of G has* Inf $=: G^-$ *and* Sup $=: G^+$.

Proof. If A and A' are rings of two prolongation, then AA' is clearly closed under comultiplication, and thus a prolongation greater than each; Hence under such a operation, we can get the minimal prolongation since the prolongation corresponds the finite etale algebra which has maximal order. The Sup follows the Cartier duality. □

Remark 2.2. We say G has UP iff $G^+ \cong G^-$.

2.1 Subgroups and quotients

Proposition 2.3. *If G_0' and G_0'' satisfy UP then G_0 has UP.*

Proof. Suppose a short exact sequence of group schemes

$$0 \longrightarrow G_0' \longrightarrow G_0 \longrightarrow G_0'' \longrightarrow 0$$

and G_0 admits a prolongation G. Then the schemetic-theoretic closure G' of G_0' in G is a prolongation of G_0', so $G'' := G/G'$ is the prolongation of G_0''. Furthermore, if H is a 2nd prolongation of G_0, which has a map $G \to H$, then we have following commutative diagram,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & G' & \longrightarrow & G & \longrightarrow & G'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f''} & & \\ 0 & \longrightarrow & H' & \longrightarrow & H & \longrightarrow & H'' & \longrightarrow & 0. \end{array}$$

If f' and f'' are isomorphisms, then f is an isomorphism by Five lemma. □

Now we can assume our group scheme G_0/K is a simple group scheme.

Definition 2.4. A $\mathbb{F}$-*module scheme* is a group scheme G equipped with a ring homomorphism $\mathbb{F} \to \mathrm{End}(G)$. $G = \mathrm{Spec}\, A$, we write $[t] : A \to A$, for $t \in \mathbb{F}$. A *Raynaud $\mathbb{F}$-module scheme* G is an $\mathbb{F}$-module scheme of the same order as $\mathbb{F}$.

Proposition 2.5. *Suppose k is algebraically closed, and G_0/K is simple of p-power order, then G_0 is canonically a Raynaud $\mathbb{F}$-module scheme, it is an $\mathbb{F}$-module scheme such that $G(\overline{K})$ is one-dimensional over $\mathbb{F}$.*

Proof. Let $V = G_0(\overline{K})$ be an irreducible $\mathbb{F}_p$-representation of the Galois group. Then $\mathbb{F} = \mathrm{End}_{Gal(\overline{K}/K)}(V)$ is a finite extension of $\mathbb{F}_p$.

Suppose k is algebraically closed, we have the exact seuqence

$$0 \longrightarrow I_K \longrightarrow Gal(\overline{K}/K) \longrightarrow Gal(\overline{k}/k) \longrightarrow 0.$$

So the $I_K \cong G_K$, and we also have a sequence of inertia group I_K,

$$0 \longrightarrow I^w \longrightarrow I_K \longrightarrow I^t \longrightarrow 0.$$

The wilde inertia group I^w is pro-p. The action of I^w on V must fix a non-zero vector in V, hence V^{I^w} is subrepresentation of V. But V is simple, so I^w acts trivially on V. Thus the action of G_K factors through I^t. Since V is an absolutely irreducible representation of an abelian group, so $\dim_{\mathbb{F}}(V) = 1$. □

Proposition 2.6. *Suppose k is algebraically closed and G_0/K is simple of p-power order, then G_0 is canonically a Raynaud $\mathbb{F}$-module scheme.*

Proposition 2.7. *Suppose UP holds for every Raynaud $\mathbb{F}$-module scheme over K^{ur}. Then UP holds for all finite order group schemes over K.*

Proof. Let G_0/K, assume k is algebraically closed. We can check that UP holds for G_0 over K^{ur} since $K \to K^{ur}$ is faithfully flat, so prolongation can be descent. Furthermore, it is known for the outer groups in an extension by Prop 2.3, so it suffices to check the case when G_0 is simple. If G_0 is prime-to p, then UP holds as prolongation is etale. Otherwise, if G_0 is p-power order, then it is a Raynaud $\mathbb{F}$-module scheme. □

Now we assume the simplest situation $\mathbb{F} := \mathbb{F}_p$, and contains $p-1$ roots of unity

2.2 Initial analysis on Raynaud $\mathbb{F}$-module scheme

We say a character $\chi : \mathbb{F}^\times \to R^\times$ is a fundamental character if the composition map $\overline{\chi} : \mathbb{F}^\times \to k^\times$ can be uniquely lifted to χ, and can be extended to an embedding of fields.

$$\begin{array}{ccc} \mathbb{F}^\times & \xrightarrow{\chi} & R^\times \\ & \searrow^{\overline{\chi}} & \downarrow \\ & & k^\times \end{array}$$

An arbitrary character $\mu : \mathbb{F}^\times \to R^\times$ can be expressed as.

$$\mu = \chi^a \qquad a \in \{0, ..., p-1\}.$$

Let $G = \operatorname{Spec} A$ be a Raynaud $\mathbb{F}$-module scheme over R. Then $G_{\overline{K}}$ is isomorphic to the constant group scheme $\mathbb{F}$ over $\overline{K}$.

Take $\epsilon_\mu : \mathbb{F} \to K$ to be an extended function of any character $\mu : \mathbb{F}^\times \to K^\times$ by $\epsilon_\mu(0) = 0$.By fixing one isomorphism of algebra $A_{\overline{K}} \to \operatorname{Hom}(\mathbb{F}, \overline{K})$, that is fixing the basis. This is an element of $A_{\overline{K}}$.

Now Let I be the augmentation ideal of A i.e.

$$I := \ker(A \longrightarrow R),$$

with the decomposition

$$I = \bigoplus_{1 \le a \le p-1} I_{\chi^a}.$$

where

$$I_\chi := \{x \in I : [t]x = \chi(t)x \text{ for all } t \in \mathbb{F}^\times\}.$$

Since $(I_\chi)_{\overline{K}}$ is generated by ϵ that I_χ is a free R-module of rank 1, so the I is rank $p-1$

Let generator $X \in I_\chi$ and some $\delta \in R$, then $X = c\epsilon$, and the relation

$$X^p = \delta X,$$

with $\epsilon^p = \epsilon$ implies

$$\delta = c^{p-1}.$$

Consider the Cartier dual $G^\vee := \operatorname{Spec} B$ of the G, where

$$B := \operatorname{Hom}_R(A, R).$$

We process the similar argument as above, the $J_{\overline{K}}$ is the augmentation ideal of $B \to \overline{K}$, and $J_{\overline{K}}$ is generated by single element

$$e = \frac{1}{p-1} \sum_{t \in \mathbb{F}^\times} \chi^{-1}(t)\{t\}.$$

Here $\{t\}$ is element of B under a fixed isomorphism to $\overline{K}[\mathbb{F}]$ corresponding to $t \in \mathbb{F}^\times$.

Note that $J_{\overline{K}}$ and $I_{\overline{K}}$ are dual under the evaluation pairing

$$\begin{aligned}\langle , \rangle : A_{\overline{K}} \times B_{\overline{K}} &\longrightarrow \overline{K}. \\ (f, \{t\}) &\longmapsto f(t),\end{aligned}$$

with bases ϵ and e are dual basis, i.e

$$\langle \epsilon, e \rangle = \frac{1}{p-1} \sum_{t \in \mathbb{F}^\times} \mu^{-1}(t) \langle \epsilon, \{t\} \rangle = \frac{1}{p-1} \sum_{t \in \mathbb{F}^\times} (\chi \mu^{-1})(t)$$

We know that $\chi^{-1}\mu =: \chi'$ is again a character, and

$$\sum_{t \in \mathbb{F}^\times} \chi'(t) = p - 1$$

if $\chi' = 1$, otherwise the sum is 0, then $\langle \epsilon, e \rangle = \delta_{\chi,\mu}$.

Now Consider the augmentation map of $B_k := k[\mathbb{F}] \to k$, the kernel J_k is spanned by the element

$$\{x\} - \{0\} =: \langle x \rangle.$$

The elements of $J_k / J_k^2 \cong k \cdot e$, satisfies the relation

$$\langle x + y \rangle = \langle x \rangle + \langle y \rangle.$$

The ideal J is also generated by these elements by Nakayama lemma.

we define two numbers, $w_a := \langle X^a, Y^a \rangle$ and $w = \langle X^p, Y^p \rangle$. These two numbers are independent on choice of prolongation G.

Proposition 2.8. *We have $w_a = a! \bmod p$, and $w = -p \bmod p^2$.*

Proof. Since each $X = c\epsilon \in I$ with $c \in \overline{K}^\times$, wlog, let $Y = c^{-1}e$, then

$$w_a = \langle \epsilon^a, e^a \rangle.$$

Since these number are independent of choice of G, so we consider the simplest case, that is $G = \mathbb{F}$, the constant group scheme over R. Hence $\epsilon \in A$ and $e \in B$.

Consider the a-comultiplicative morphism

$$\Delta_a : A \longrightarrow A^{\otimes a}.$$

Then we can write

$$w_a = \langle (\Delta_a(\epsilon))^a, e^{\otimes a} \rangle$$

Each $\Delta_a(\epsilon)$ is

$$\underbrace{\epsilon \otimes 1 \otimes \cdots \otimes 1 + 1 \otimes \epsilon \otimes \cdots \otimes 1 + \cdots + 1 \otimes \cdots \otimes \epsilon}_{a\text{-terms}} \bmod p.$$

Therefore, we get

$$\langle \epsilon^a, e^a \rangle = \langle \Delta_a(\epsilon)^a, e^{\otimes a} \rangle \equiv a! \bmod p,$$

since the coefficient of $\epsilon^{\otimes a}$ is $a!$. Hence we obtain

$$w_a = a! \mod p.$$

Furthermore, take

$$w = \langle \epsilon^p, e^p \rangle = \langle \Delta_p(\epsilon)^p, e^p \rangle = \langle (x+y)^p, e^p \rangle$$

here take $y \in pA^{\otimes p}$ and x be the remaining part. So we get

$$(x+y)^p \equiv x^p \mod p^2.$$

Hence, $w \equiv p! \equiv -p \mod p^2$.

□

3 The Unique Prolongation

Theorem 3.1 (Structure)**.** *Any prolongation $G = \operatorname{Spec} A$, is in the form $A \cong R[X]/(X^p - \delta X)$, with $v(\delta) \leq e$.*

Proof. Since X^a span I and Y^a span J, and the pairing $\langle X^a, Y^a \rangle = w_a$, from last section, we get

$$w_a \equiv a! \bmod p.$$

Since each $a \leq p-1$, and each for each a satisfies $p \nmid a$, this implies $w_a \in R^\times$.

Furthermore, there are $p-1$ numbers of character μ on $\mathbb{F}^\times$. And the algebra A is rank p over R. Thus,

$$I = \bigoplus_{a=1}^{p-1} R \cdot X^a.$$

So we get A is generated by X, with relation $X^p = \delta X$.

On the other hand

$$w = \delta\gamma \equiv -p \bmod p^2,$$

hence

$$v(w) = v(p) = e = v(\delta) + v(\gamma).$$

We get

$$v(\delta) \leq e.$$

□

Theorem 3.2 (Existence)**.** *Let $\delta \in R$ such that $v(\delta) \leq e$, and A be the quotient,*

$$A := R[X]/(X^p - \delta X).$$

Then there exists a unique structure of a Raynaud $\mathbb{F}$-module scheme on G, such that $[t]X = \chi(t)X$ for all $t \in \mathbb{F}^\times$.

Proof. Let $X = c\epsilon$, where $\epsilon \in I_{\overline{K}}$, $c \in \overline{K}$, then

$$X^p = (c\epsilon)^p = c^p \epsilon^p = \delta c\epsilon = \delta X$$

with the relation of $\delta = c^{p-1}$ and valuation condition, we obtain the

$$\epsilon^p = \epsilon.$$

Which gives the group scheme

$$\operatorname{Spec} \overline{K}[T]/(T^p - T).$$

But this is isomorphic to $\mathbb{F}$. Consider the Cartier dual $\operatorname{Spec} B$ here, we see the valuation condition is that $v(\delta) \leq e$ this implies $v(\gamma) \geq 0$, so $\gamma \in R$, which shows that the Cartier dual B is that closed under multiplication. Hence the algebra A closed under comultiplication which gives the $G := \operatorname{Spec} A$ group scheme structure.

We see that for any $t \in \mathbb{F}^\times$, the action preserves the relation $X^p - \delta X$, i.e.

$$[t](X^p - \delta X) = [t]X^p - [t]\delta X = \chi(t)X^p - \chi(t)\delta X = \chi(t)(X^p - \delta X).$$

The Raynaud $\mathbb{F}$-module structure on G comes from some choice of $c \in \overline{K}^\times$ such that $\delta = c^{p-1}$, but such a choice is uniquely determined by a single $p-1$ roots of unity.

Which is independent of the choice of c. Hence the Raynaud $\mathbb{F}$-module structure on G is unique. □

The following proposition shows that the homomorphism of different prolongation depends on the element $a \in R$.

Proposition 3.3. *The set of $\mathbb{F}$-module homomorphism maps*

$$f : G_\delta \longrightarrow G_{\delta'}$$

correspond to $a \in R$ such that

$$a\delta' = a^p\delta.$$

The number a corresponds to the map f which is given on rings by

$$X' \longmapsto aX.$$

Proof. • The Forward direction:

Let $f : G_\delta \to G_{\delta'}$, the morphism of the group scheme corresponds to the morphism of hopf-algebra,

$$f^* : A_{\delta'} \longrightarrow A_\delta.$$

Since the $G_\delta, G_{\delta'}$ have Raynaud $\mathbb{F}$-module scheme structure, and f^* commutes with the $\mathbb{F}$-action,

$$f^*([t]X') = f^*(\chi(t)X) = [t]f^*(X') = \chi(t)f^*(X').$$

Then $f^*(X') \in A_\delta$, but A_χ is 1-dim space. So we get

$$f^*(X') = aX$$

for some $a \in R$. Hence, we have the following identifications:

$$a^p\delta X = a^pX^p = f^*(X')^p = f^*((X')^p) = f^*(\delta' X') \overset{\delta' \in R}{=} \delta' f^*(X') = \delta' aX$$

Therefore, we have the relation

$$a^p\delta = a\delta'$$

- The Backward direction: Given a morphism on rings $R[X'] \to A_\delta$ by

$$X' \longmapsto aX$$

and suppose the number a satisfying

$$a^p\delta = a\delta'.$$

We only need to check that the relation,

$$(X')^p - \delta' X'.$$

of $A_{\delta'}$ vanish.

The image of this relation is:

$$(aX)^p - \delta' aX = a^pX^p - \delta' aX = (a^p\delta - a\delta')X.$$

By our assumption this is zero. Therefore the map gives

$$f^* : A_{\delta'} \to A_\delta.$$

So we get:

$$f : G_\delta \to G_{\delta'}.$$

□

Proposition 3.4. *Suppose $e < p-1$, and $f : G \to G'$ is a map of Raynaud $\mathbb{F}$-module scheme over R which induces an isomorphism over K. Then f is an isomorphism.*

Proof. Let $G = G_\delta, G' = G_{\delta'}$, so f corresponds to a number a with relation $a\delta' = a^p\delta$. Then let $v(a)$ to be the maxiaml, we get

$$pv(a) \leq v(a^p\delta) = v(a\delta') \leq v(a) + e,$$

The last inequality is due to Theorem 3.1. Hence

$$(p-1)v(a) \leq e.$$

This forces $v(a) = 0 \implies a \in R^\times$. So f is an isomorphism. □

Proof of Raynaud's theorem. We can apply the proposition to $f : G^+ \to G^-$ under hypothesis $e < p-1$, so G has UP. □

References

[Sno13] Andrew Snowden. Course on mazur's theorem. Lecture notes for Math 679: Elliptic Curves, University of Michigan, 2013. Fall 2013 course notes.